LUYI TONG

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EDUCATION

University of Washington | Seattle, United States
Master of Science, Biostatistics | Sept 2024 – Mar 2026 (Expected)

McGill University | Montreal, Canada
Bachelor of Arts, Honours Probability and Statistics | Sept 2020 - June 2024
Minor: Economics | GPA: 3.72/4.00

CORE SKILLS

- Python, Java, R, SQL, PowerBI / Tableau
- Survival Analysis, Causal Inference, Longitudinal Modeling, Logistic & Cox Regression, Model Validation

EXPERIENCE

Data Analyst & Biostatistics Intern
Tango Technologies LLC dba Future Force — Seattle, United States *Jun. 2025 – Sept. 2025*

- Applied clustering and statistical inference methods to structured and unstructured datasets to identify opportunity patterns, improving identification coverage by 20% and enhancing ranking accuracy
- Developed predictive scoring models using machine learning methods, achieving 95% classification accuracy and supporting prioritization of high-impact interventions
- Conducted user behavior analyses for Eldercare AI using SQL and Python, identifying caregiver burden indicators and increasing user engagement by 10%
- Built predictive risk models to identify high-risk eldercare scenarios, improving early identification performance by 15%
- Designed dashboards to communicate analytic findings and collaborated with cross-functional teams to translate quantitative results into product and intervention improvements

Digital Marketing & Data Analytics Intern
Ningbo Chunqiu Guantu Culture and Media Co., Ltd — Zhejiang, China

- Developed large-scale data pipelines integrating 20+ cross-channel data sources, enabling user-level attribution across impressions, clicks, and conversions
- Implemented Shapley-value–based attribution models to optimize budget allocation, increasing campaign ROI by 25%
- Designed segmentation models (RFM + cross-category) to identify key customer groups, improving retention among high-value users by 15%
- Conducted Monte Carlo simulations for GMV forecasting, reducing planning cycles from 7 days to 72 hours and improving operational decision-making
- Built dashboards to monitor key performance indicators and support data-driven strategy adjustments

PROJECT WORK

Cancer Treatment Outcomes Analysis (Inflammatory Breast Cancer) *[Ongoing]*
Technologies: R (Survival Analysis, MatchIt, survey), SQL, Data Visualization

- Constructed analysis-ready cohort by integrating SEER registry data (50k+ records) with clinical datasets
- Designed causal inference framework using propensity score matching, weighting, and stratification to estimate treatment effects in non-randomized settings
- Implemented Cox proportional hazards models and survival analyses to evaluate overall survival and time-to-event outcomes
- Conducted sensitivity and robustness analyses across matching specifications and model forms to assess stability of findings
- Produced summaries and visualizations to communicate treatment effectiveness results

Eldercare AI Analytics
Technologies: Python (Pandas, Scikit-learn, NumPy, Matplotlib), SQL, Clustering

- Analyzed caregiver and patient interaction data to identify behavioral indicators of care burden
- Built predictive risk scoring models to identify high-risk eldercare scenarios
- Applied clustering methods to identify heterogeneous caregiver subgroups and support targeted interventions
- Translated analytical findings into actionable design insights for AI-based care tools

Bayesian Mendelian Randomization for Cardiovascular Disease
Technologies: R, Bayesian Modeling, Instrumental Variable Methods

- Investigated causal effects of cardiometabolic risk factors on cardiovascular disease outcomes using Mendelian randomization with genetic instruments
- Applied Bayesian MR models to estimate exposure–outcome relationships using clinical and genetic data, addressing confounding and reverse causation in observational settings
- Assessed robustness of causal estimates under alternative assumptions on instrument validity and pleiotropy
- Compared Bayesian and conventional MR approaches to evaluate sensitivity to weak instruments and model misspecification
- Summarized findings on the causal role of key risk factors in cardiovascular disease through technical reports and presentations